

SYED MUHAMMAD HAMZAH RIZVI

CONTROLS & PROCESS ENGINEER | AUTOMATION | PROCESS IMPROVEMENT | DIGITAL ENABLEMENT

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PROFILE

Control and process engineer with an MSc in Robotics and Automation (Distinction) and hands-on experience across industrial automation, commissioning and digital manufacturing. Led implementation of a building-management system for utility monitoring and automation that contributed to an approximately 25% improvement in energy efficiency, and established an in-house electrical repair capability that reduced reliance on external support. Skilled in PLC, VFD and HMI programming, Python, manufacturing analytics and dashboard development, with current experience delivering API-integrated digital systems into live production environments across 10+ countries. Strong interest in Industry 4.0, process optimisation and continuous improvement within pilot plant and manufacturing environments.

CORE SKILLS ALIGNED TO ROLE

Process, pilot plant & manufacturing: Manufacturing support, process optimisation, equipment installation and commissioning, operational readiness, troubleshooting, root-cause analysis, maintenance support, electrical drawings, wiring diagrams, motors, drives, sensors, safety checks.

Digital enablement & data workflows: Workflow digitisation, data capture workflows, REST APIs, JSON workflows, webhooks, authentication, Python, C++, system integration, data quality, production behaviour testing, technical documentation.

Control, automation & equipment: PLC, VFD and HMI design/troubleshooting, control systems, industrial automation, BMS implementation, utility monitoring, electrical panels, Omron robot control and command, system commissioning.

Delivery, governance & adoption: Stakeholder management, requirements gathering, implementation planning, risk-aware change, cross-functional coordination, escalation management, handover, training materials, knowledge transfer, SOP-style documentation.

Relevant Software: Power BI, Excel. HTML/CSS

PROFESSIONAL EXPERIENCE

INNOVATIVE TECHNOLOGY - TECHNICAL SOFTWARE CONSULTANT | Manchester, UK | March 2023 - Present

- Lead customer-facing deployment and integration of AI-powered biometric and identity verification systems across international client environments, coordinating customer engineering teams, internal specialists and support functions to achieve reliable go-live.
- Translate technical and operational requirements into implementation plans, data/API workflows, validation steps, handover documentation and user adoption support.
- Diagnose complex deployment issues spanning software, hardware devices, APIs, mobile applications and networks; isolate root causes and coordinate corrective action with internal and external stakeholders.
- Design, debug and support REST API integrations, JSON workflows, authentication processes and webhooks, with focus on data quality, traceability and stable production behaviour.
- Act as escalation point for deployment-critical failures, managing multiple parallel priorities while maintaining clear communication, detail control and delivery discipline.
- Produce technical documentation, integration guides, workflow diagrams, training material and handover packs to reduce implementation risk and support repeatable adoption.

TRIPACK FILMS LTD. - AUTOMATION ENGINEER | Pakistan | March 2020 - September 2021

- Supported automation, electrical and reliability work across a large-scale manufacturing site, collaborating with operations, maintenance, projects and engineering teams to improve production equipment performance.
- Installed, configured, tested and troubleshoot automation systems, sensors, electrical panels, motors, drives and control-related equipment in a live production environment.
- Supported equipment commissioning and production readiness activities, including electrical configuration, safety checks, system integration, field troubleshooting and handover.
- Conducted fault-finding across electrical, mechanical and control-system interfaces; used root-cause analysis to identify practical improvements and reduce repeated downtime issues.
- Interpreted electrical drawings, wiring diagrams and technical documentation to support maintenance, equipment modifications and safe issue resolution.
- Worked with technicians and engineers to standardise troubleshooting approaches, improve internal capability and reduce dependency on external repair support.

EVOLLIGENCE - SOFTWARE CONSULTANT | Pakistan | December 2019 - March 2020

- Built and supported backend application logic using structured APIs and data-driven workflows, collaborating with development teams to deliver against requirements.

SELECTED PROJECTS

MANUFACTURING ANALYTICS DASHBOARD | PERSONAL PROJECT - github.com/hamzahrizvi/manufacturing-dashboard

- Designed and developed an interactive manufacturing dashboard to monitor production performance, equipment status and operational KPIs.

- Built real-time visualisations for production output, quality metrics, machine uptime, energy consumption and maintenance scheduling.
- Implemented data-driven dashboards to support process monitoring, trend analysis and operational decision-making.
- Developed responsive web interfaces using modern front-end technologies with a Python-based backend for manufacturing data management.
- Applied digital manufacturing principles to improve process visibility and support continuous improvement initiatives.
- Demonstrated knowledge of Industry 4.0 concepts including digital monitoring, KPI reporting and manufacturing analytics.

METALLIZING MACHINE INSTALLATION - BOBST MANCHESTER

- Supported installation and commissioning of industrial metallizing machinery within a live manufacturing environment.
- Assisted with electrical configuration, safety checks, system integration, troubleshooting and equipment handover.
- Worked with engineering teams to resolve installation issues and confirm readiness for production use.

BUILDING MANAGEMENT SYSTEM IMPLEMENTATION

- Contributed to BMS implementation for utility monitoring, automation, data collection and energy-efficiency improvement.
- Supported system configuration and optimisation of building utility processes using improved monitoring and control.
- Helped achieve approximately 25% improvement in energy efficiency through better monitoring and automation.

ELECTRICAL REPAIR & TRAINING INITIATIVE

- Set up an in-house repair workbench for electrical components, motors and production equipment.
- Trained technicians on basic troubleshooting, repair procedures and safe handling of electrical equipment.
- Improved internal fault-resolution capability and reduced reliance on external repair support.

EDUCATION

MSc Robotics and Automation - University of Salford, UK | 2021 - 2022 | Graduated with Distinction | Minor in AI

BEng Electrical Engineering - National University of Science and Tech, Pakistan | 2015 - 2019 | Minor in Robotics and Automation

CERTIFICATIONS

- PLC, VFD & HMI Design, Programming and Troubleshooting
- Omron Robot Control and Command